

Lecture 11

Calogero–Moser: the model, its Lax pair and its scattering

Objectives. Every particle now interacts with every other, through an inverse-square potential or one of its three relatives. We derive Moser’s Lax pair for three of the four from one functional identity (the fourth, elliptic, solves the same identity, cited); prove the integrals independent; and find that involution needs an r -matrix of a kind [Lecture 8](#) only warned about. It depends on the positions and is not skew, and for three particles no r at all puts the bracket in [Lecture 8](#)’s single-commutator form. The involution theorem survives anyway. The rational gas scatters like the Toda chain, momenta permuted, but with no displacement whatever; in a harmonic trap every orbit is periodic.

11.1 The four pair potentials

N unit masses on a line, positions $q_1, \dots, q_N$, interacting in pairs:

$$H = \frac{1}{2} \sum_{j=1}^N p_j^2 + g^2 \sum_{j < k} V(q_j - q_k), \quad g > 0. \quad (11.1)$$

Four even pair potentials make this integrable ([Figure 11.1](#)):

model	$V(x)$	motion
rational	$1/x^2$	scattering
hyperbolic	$1/\sinh^2 x$	scattering, short range
trigonometric	$1/\sin^2 x$	on a circle, confined
elliptic	$\wp(x)$	on a circle, confined

A length scale a can be restored as $a^2 V(ax)$; we set $a = 1$. The *Weierstrass function* is, for a lattice $\Lambda = 2\omega_1\mathbb{Z} + 2\omega_2\mathbb{Z}$ in $\mathbb{C}$ — here always rectangular, with ω_1 real and ω_2 purely imaginary, so that $\wp$ is real on the real line with real period $2\omega_1$ —

$$\wp(x) = \frac{1}{x^2} + \sum_{w \in \Lambda \setminus \{0\}} \left(\frac{1}{(x-w)^2} - \frac{1}{w^2} \right),$$

even, *doubly periodic* ($\wp(x + 2\omega_1) = \wp(x + 2\omega_2) = \wp(x)$), with a double pole at every lattice point. We use nothing else about it. Sending one period to infinity turns it, up to scale and an additive constant, into $1/\sin^2$ or $1/\sinh^2$; sending both gives $1/x^2$ (Perelomov, §3.1). The elliptic model contains the other three as limits. The list rational / trigonometric (with hyperbolic) / elliptic is the list of r -matrices met in § 8.7; that it recurs is not a coincidence, but this course does not prove it.

Every V has a double pole at $x = 0$ and is repulsive near it, so two particles never meet and their order is preserved. What happens far apart splits the list in two (Figure 11.2). For $1/x^2$ and $1/\sinh^2 x$ the potential dies away and the particles fly apart: *scattering*, as for open Toda. The phase space is the chamber $q_1 < \dots < q_N$ times $\mathbb{R}^N$. For $1/\sin^2 x$ and $\wp$ the potential is periodic, with period $d = \pi$ or $2\omega_1$, so the positions matter only modulo d : N particles on a circle of length d , each hemmed in by its neighbours, *confined*. The cyclic order of the particles is preserved; in each fixed cyclic order a configuration is a point of the circle (where particle 1 is) and N positive gaps adding up to d (Perelomov, (3.1.16)), so that component of phase space is a circle times an open $(N - 1)$ -simplex, times $\mathbb{R}^N$ for the momenta. We work with the rational model and say at each step what carries over. (One generalisation, cited: the differences $q_j - q_k$ are the values at q of the roots $e_j - e_k$ of the Lie algebra $\mathfrak{sl}_N$, and summing the same potential over the roots of any other root system, with suitable couplings, keeps the model integrable — Olshanetsky and Perelomov 1976. Lecture 12 shows where $\mathfrak{sl}_N$ comes from.)

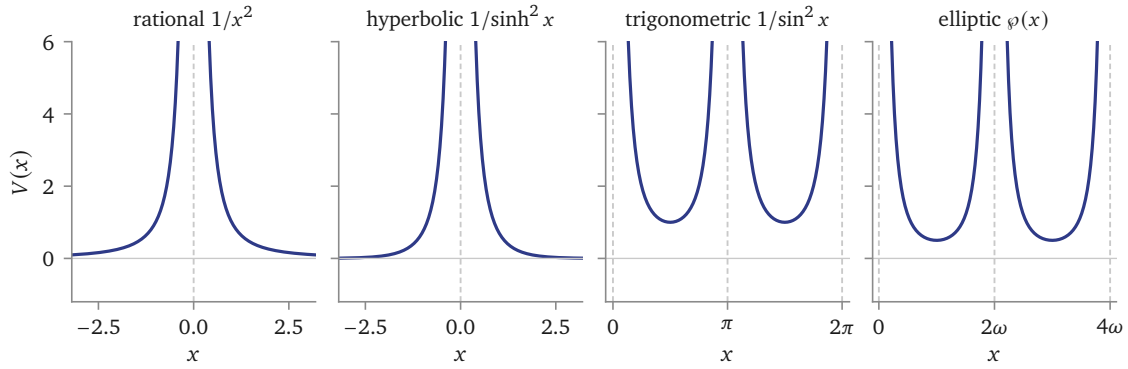


Figure 11.1: The four pair potentials. Each has a double pole at $x = 0$, hence impenetrable particles. The first two decay (scattering); the last two are periodic (particles on a circle). For $\wp$ the lattice is rectangular, with real half-period ω (scripts/figures/cm_potentials.py, which checks $\wp(x) = x^{-2} + O(x^2)$, the period and Weierstrass's equation $\wp'^2 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3)$).

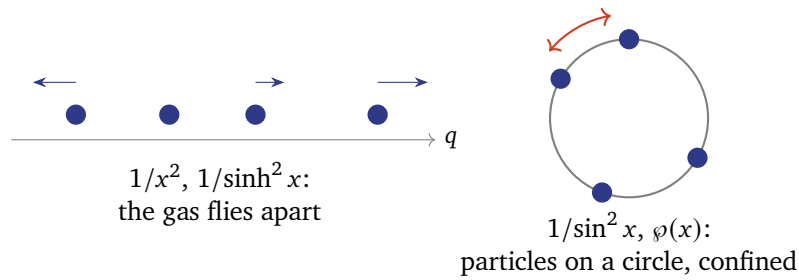


Figure 11.2: Scattering versus confinement. On the line the repulsion eventually wins and the particles separate; on the circle each is caught between its two neighbours and the motion is bounded.

11.2 Moser’s Lax pair, with M derived

Proposition 7.8 stated Moser’s pair for the rational model and promised a derivation. Here it is, done so that the four models come out together.

Choosing L

Follow the recipe of § 7.6: find L with $\text{tr } L^2 = 2H$, then solve for M . Every pair interacts, so every off-diagonal slot is used: try

$$L_{jk} = p_j \delta_{jk} + i g x(q_j - q_k) (1 - \delta_{jk}), \quad (11.2)$$

with x a real *odd* function still to be found. Oddness makes L Hermitian, since $L_{kj} = i g x(q_k - q_j) = -i g x(q_j - q_k) = \overline{L_{jk}}$, and then

$$\text{tr } L^2 = \sum_j p_j^2 + \sum_{j \neq k} (i g)^2 x(q_{jk}) x(q_{kj}) = \sum_j p_j^2 + g^2 \sum_{j \neq k} x(q_{jk})^2, \quad q_{jk} := q_j - q_k.$$

Within this ansatz, this is $2H$ exactly when $x^2 = V$. So $x = 1/u$ for the rational model, and likewise $1/\sin u$ and $1/\sinh u$ (Figure 11.3).

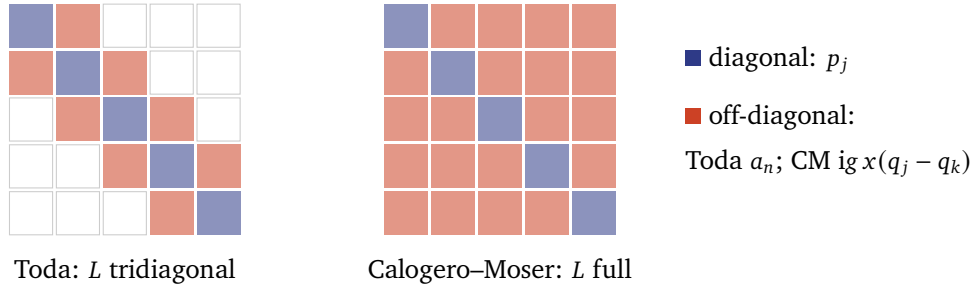


Figure 11.3: The shape of the Lax matrix follows the interaction. Toda’s neighbours give a tridiagonal L ; in Calogero–Moser every pair interacts, and every off-diagonal slot is filled.

Solving for M

Along the flow, $\dot{L}_{jj} = \dot{p}_j$ and, for $j \neq k$, $\dot{L}_{jk} = i g x'(q_{jk})(p_j - p_k)$. We look for M with $[M, L]$ equal to this. M is never unique — adding any function of L changes nothing (M is determined only up to the commutant of L , Lecture 7) — so we make an ansatz, the natural one for a pairwise interaction: M is *anti-Hermitian*, depends on the positions only, and M_{jk} ($j \neq k$) is a function of $q_j - q_k$. Within it, three observations fix M .

The part linear in p . Off the diagonal, the only terms of $[M, L]_{jk}$ containing momenta then come from the diagonal of L : $M_{jk}L_{kk} - L_{jj}M_{jk} = M_{jk}(p_k - p_j)$. Matching $\dot{L}_{jk}$ forces

$$M_{jk} = -i g x'(q_{jk}) \quad (j \neq k), \quad (11.3)$$

anti-Hermitian because x' is even.

The diagonal. $[M, L]_{jj}$ involves only off-diagonal entries:

$$[M, L]_{jj} = \sum_{l \neq j} (M_{jl}L_{lj} - L_{jl}M_{lj}) = -g^2 \sum_{l \neq j} 2x(q_{jl})x'(q_{jl}) = -g^2 \sum_{l \neq j} V'(q_{jl}) = \dot{p}_j,$$

using oddness twice and $V' = 2xx'$. So the diagonal of the Lax equation is Newton's law, whatever the diagonal of M is.

The rest. What remains of $[M, L]_{jk}$, $j \neq k$, must vanish: the terms through a third index l , and the diagonal of M ,

$$(\mathrm{i}g)^2 \sum_{l \neq j, k} (x(a)x'(b) - x'(a)x(b)) + \mathrm{i}g x(a+b) (M_{jj} - M_{kk}) = 0, \quad a = q_{jl}, \quad b = q_{lk}, \quad a+b = q_{jk}.$$

This is where the model is decided. Suppose x satisfies the *functional identity*

$$x(a)x'(b) - x'(a)x(b) = x(a+b)(V(a) - V(b)). \quad (11.4)$$

Then the condition becomes $M_{jj} - M_{kk} = -\mathrm{i}g \sum_{l \neq j, k} (V(q_{jl}) - V(q_{kl}))$, and since V is even the two missing terms $l = k$ in the first sum and $l = j$ in the second would be equal, so it is solved by

$$M_{jj} = -\mathrm{i}g \sum_{l \neq j} V(q_{jl}), \quad (11.5)$$

and, within the ansatz, only by this up to adding a common function to every M_{jj} — a multiple of 1 , which commutes with everything.

Theorem 11.1 (Moser's Lax pair, derived). *Let $x(u)$ be $1/u$, $1/\sin u$ or $1/\sinh u$, and $V = x^2$. With L as in (11.2) and M as in (11.3)–(11.5), the $\dot{L} = [M, L]$ along every solution of (11.1), for every N . Conversely, for $x = 1/u$ and $1/\sinh u$, the Lax equation gives Newton's law and $\dot{q}_j - \dot{q}_k = p_j - p_k$: the equations of motion up to a common drift of all positions, which L cannot see. L is Hermitian, M anti-Hermitian, $\mathrm{tr} L^2 = 2H$.*

Proof. It remains to prove (11.4). Write $x = 1/s$ with $s(u) = u$, $\sin u$ or $\sinh u$. Then $x' = -s'/s^2$, and (11.4) becomes, after multiplying by $s(a)^2 s(b)^2$,

$$s'(a)s(b) - s(a)s'(b) = \frac{s(b)^2 - s(a)^2}{s(a+b)}.$$

For all three the left side is $s(b-a)$ (for $\sin$: $\cos a \sin b - \sin a \cos b$, and $s(b)^2 - s(a)^2 = s(b+a)s(b-a)$ ($b^2 - a^2 = (b+a)(b-a)$, and the product formulas for $\sin$ and $\sinh$). The three observations above then give the Lax equation entry by entry. Conversely, its diagonal is Newton's law, and its off-diagonal entries say $x'(q_{jk})(\dot{q}_j - \dot{q}_k) = x'(q_{jk})(p_j - p_k)$, and x' never vanishes for $1/u$ and $1/\sinh u$. (For $1/\sin u$ it vanishes at $u = \pm\pi/2$, and there the Lax equation alone does not recover $\dot{q}_j$.) ■

Example 11.2 (Two particles, by hand). For $N = 2$, $x = q_1 - q_2$ and the rational model,

$$L = \begin{pmatrix} p_1 & \mathrm{i}g/x \\ -\mathrm{i}g/x & p_2 \end{pmatrix}, \quad M = \frac{\mathrm{i}g}{x^2} \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}, \quad [M, L] = \begin{pmatrix} 2g^2/x^3 & \mathrm{i}g(p_2 - p_1)/x^2 \\ \mathrm{i}g(p_1 - p_2)/x^2 & -2g^2/x^3 \end{pmatrix}.$$

With $H = \frac{1}{2}(p_1^2 + p_2^2) + g^2/x^2$, the diagonal is $\dot{p}_1 = -\partial H/\partial q_1 = 2g^2/x^3$ and $\dot{p}_2 = -\dot{p}_1$, and the off-diagonal is $\frac{d}{dt}(\mathrm{i}g/x) = -\mathrm{i}g \dot{x}/x^2$ with $\dot{x} = p_1 - p_2$. There is no third index, so the functional identity is not used: it is what makes $N \geq 3$ work.

For $N = 3$ and all three choices, and for the rational model up to $N = 4$, `scripts/check-111-algebra.py` checks the Lax equation symbolically, and checks that the diagonal of M is forced up to $c1$.

Proposition 7.8 is the case $x = 1/u$: its M was not guessed after all.

The *elliptic model*. The identity (11.4) with V replaced by $\wp$ plus a constant also has odd solutions, built from Jacobi's elliptic functions, for instance $x = \mathrm{cn} u / \mathrm{sn} u$ (Calogero 1975;

Perelomov, §3.1, eq. 3.1.12) · *cited*. So all four models have Moser-type Lax pairs without a spectral parameter. Krichever (1980) found the elliptic pair *with* one, the version that carries a spectral curve (Babelon–Bernard–Talon, §7.4); we do not need it.

11.3 The integrals $\text{tr } L^k$, and independence

By [Theorem 7.2](#) the N functions $F_k = \text{tr } L^k$ are conserved, and since L is Hermitian they are real. The first three are

$$F_1 = \sum_j p_j, \quad F_2 = 2H, \quad F_3 = \sum_j p_j^3 + 3g^2 \sum_{j < k} (p_j + p_k) V(q_{jk}) : \quad (11.6)$$

total momentum, energy, and a cubic integral with the same shape as the Toda chain's (7.14), the neighbours replaced by all pairs.

Theorem 11.3 (Independence). *For $x = 1/u$, $1/\sin u$ or $1/\sinh u$, $F_1, \dots, F_N$ are independent on a dense open subset of the phase space. Together with [Theorem 11.6](#) below, the rational Calogero–Moser system satisfies [Definition 4.6](#).*

Proof. Expand $\text{tr } L^k$ as a sum over closed walks $j_1 \rightarrow j_2 \rightarrow \dots \rightarrow j_k \rightarrow j_1$ through the indices, each step contributing $L_{j_i j_{i+1}}$. A step that stays put contributes a momentum; a step that moves contributes no momentum, and a closed walk that moves at all moves at least twice. So $F_k = \sum_j p_j^k + R_k$ with R_k of degree at most $k - 2$ in the momenta. In the Jacobian $\det(\partial F_k / \partial p_n)$ the part of top degree in p therefore comes from $\sum_j p_j^k$ alone, and it is $\det(k p_n^{k-1}) = N! \prod_{i < j} (p_j - p_i)$, the Vandermonde determinant (§ 9.3). At fixed q the Jacobian is a non-zero polynomial in p , so, exactly as in [Theorem 10.5](#), the set where it does not vanish is open and dense, and there the dF_k are independent. ■

The Toda chain of [Lecture 9](#) had independence at every point; here, as for the ring of [Lecture 10](#), a dense open set is what the proof gives, and all that [Definition 4.6](#) asks.

11.4 Involution, and the dynamical r -matrix

Now involution. [Theorem 8.7](#) needs a Sklyanin bracket (8.7): a *numerical* matrix r with $\{L_1, L_2\} = [r, L_1 + L_2]$. The open Toda chain had one ([Proposition 9.4](#)). [Lecture 8](#)'s reading section also met a weaker, *two-term* form, which Babelon and Viallet showed always exists behind involution:

$$\{L_1, L_2\} = [r_{12}, L_1] - [r_{21}, L_2], \quad r_{21} = P r_{12} P, \quad (11.7)$$

with r allowed to depend on the point of phase space. When r is skew, $r_{21} = -r_{12}$, this is (8.7). For Calogero–Moser, the constant r of [Lecture 8](#) does not exist, and for three particles not even a dynamical one in the single-commutator form:

Proposition 11.4 (No r of Lecture 8's kind). *For the rational model:*

- (a) *no constant r satisfies either (8.7) or the two-term form (11.7), for $N = 2, 3$;*
- (b) *for $N = 3$ there is not even a dynamical r in the single-commutator form: at the point $q = (\frac{1}{3}, \frac{2}{3}, \frac{5}{3})$, $p = (\frac{1}{2}, -\frac{1}{2}, -\frac{3}{2})$, $g = \frac{1}{2}$, the linear equations for the 81 entries of r are inconsistent.*

Proof. Each statement is the inconsistency of a finite linear system, solved in exact rational arithmetic by `scripts/check-l11-algebra.py`: (a) for each form and each N , at three points at once, since a constant r must work at all of them; (b) at the one point, since a dynamical r must work at every point and so at this one. (For $N = 2$ the single-commutator system is consistent pointwise, which is why (b) says $N = 3$.) ■

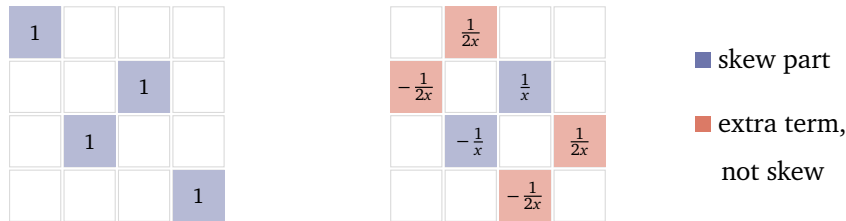
For the $N = 3$ obstruction (b) the mechanism is that of [Lecture 8](#)’s reading example. At that point, in an eigenbasis of L , the matrix $L_1 + L_2$ is diagonal with equal entries in the slots $(ij), (ji)$, so every commutator with it vanishes there, whatever r is; there $\{L_1, L_2\}$ does not. (For $N = 2$ those entries of the bracket do vanish, which is why a pointwise solution exists.) What does exist is the weaker form that the Babelon–Viallet theorem guarantees, with two independent terms, and r depending on the positions.

Proposition 11.5 (The dynamical r -matrix · cited). For the rational model, with $q_{jk} = q_j - q_k$,

$$r_{12} = \sum_{j \neq k} \frac{1}{q_{jk}} \left(E_{jk} \otimes E_{kj} + \frac{1}{2} (E_{jj} + E_{kk}) \otimes E_{jk} \right) \quad (11.8)$$

satisfies the two-term form (11.7). (Avan and Talon 1993; Babelon–Bernard–Talon, (7.25), whose rational limit is $-r_{12}$ because their bracket is $\{p, q\} = 1$. Checked symbolically for $N = 2, 3, 4$. Avan and Talon give the corresponding r for the trigonometric and hyperbolic models too, so [Theorem 11.6](#) applies there as well · cited.)

Read (11.8) against [Lecture 8](#) ([Figure 11.4](#)). The first sum is skew, $P(\cdot)P = -(\cdot)$, and looks like § 8.7’s P/λ with its diagonal removed and the spectral parameter replaced by a position difference. The second sum is the *extra term*: it is not skew, which is why the two terms of (11.7) do not combine into one commutator. (For $N = 2$ a different, single-commutator representative exists pointwise; for $N = 3$ none does, [Proposition 11.4](#).) And g has disappeared: the same r serves every coupling.



[Lecture 8](#): $r = cP/\lambda$ one numerical matrix Calogero–Moser, $N = 2$, $x = q_1 - q_2$: a matrix that moves with the particles

Figure 11.4: Two r -matrices on $\mathbb{C}^2 \otimes \mathbb{C}^2$, rows and columns in the order 11, 12, 21, 22. Left, [Lecture 8](#)’s rational one: numbers. Right, the Calogero–Moser r of (11.8) for two particles: entries that depend on the separation, and an extra term (red) that breaks the skew symmetry, which is why this r enters through the two-term form (11.7).

Theorem 11.6 (Involution). If L satisfies (11.7) for any matrix-valued functions r_{12}, r_{21} on phase space, then $\{\text{tr } L^n, \text{tr } L^m\} = 0$ for all n, m . In particular the Calogero–Moser integrals F_k are in involution.

Proof. Step 1 of [Theorem 8.7](#), the chain rule (8.10), never looked at the bracket. Substituting

(11.7),

$$\{\mathrm{tr} L^n, \mathrm{tr} L^m\} = nm \left[\mathrm{tr} (A [r_{12}, L_1]) - \mathrm{tr} (A [r_{21}, L_2]) \right], \quad A = L_1^{n-1} L_2^{m-1}.$$

A commutes with L_1 and with L_2 separately (Remark 8.2), and Step 3’s cyclicity argument kills each trace. The values of r at the point where the bracket is evaluated are all that enter; that they vary from point to point costs nothing. ■

So the machinery did not fail at its first real test. What Lecture 8 proved was more than it assumed: the involution theorem needs only that the bracket is a sum of commutators with L_1 and with L_2 , and Babelon–Viallet (Lecture 8, reading) says that some such form always exists behind involution. A constant, skew r is the exception — the open Toda chain has one — and a dynamical r the general situation, as Babelon and Viallet describe it. What *does* change is Jacobi: Theorem 8.11 was proved for a numerical skew r . For their elliptic dynamical r , Babelon–Bernard–Talon show by direct calculation a generalised Yang–Baxter equation (7.21), with new terms that are Poisson brackets of L with r , and that it implies Jacobi · *cited*; in general Jacobi gives a weaker condition, and r has gauge freedom. Here Jacobi is not in question: the bracket is the canonical one.

11.5 Scattering, and the comparison with Toda

In the rational model the particles repel and the potential dies away, so, as in § 9.7, they separate (Moser). Once they have, $L \rightarrow \mathrm{diag}(p)$, and the one-line argument after Theorem 9.9 applies word for word: *the asymptotic momenta are the eigenvalues* $\lambda_1 < \dots < \lambda_N$ of L , arriving in decreasing order along the line and leaving in increasing order (Figure 11.5). The scattering is elastic and non-diffractive.

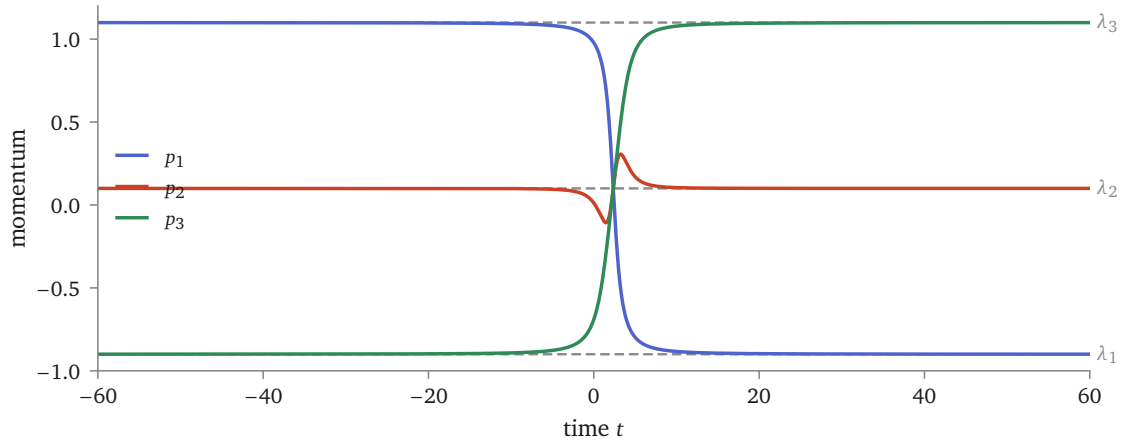


Figure 11.5: Three particles of the rational gas, $g = 1$: momenta (solid) and the eigenvalues of Moser’s L (dashed). The momenta are exchanged; the eigenvalues do not move (scripts/figures/cm_scattering.py, which asserts the spectrum and the energy constant to 10^{-9} and both asymptotic orders).

The positions are where Calogero–Moser and Toda part company.

Proposition 11.7 (Two particles). *For $N = 2$, with the gap $s = q_2 - q_1 > 0$ and relative*

speed $v = \lambda_2 - \lambda_1$, every solution has

$$s(t)^2 = v^2(t - t_0)^2 + \frac{4g^2}{v^2}, \quad \text{so} \quad s(t) = v|t - t_0| + O(1/t). \quad (11.9)$$

The incoming straight lines continue into the outgoing ones with no displacement.

Proof. The gap obeys $\ddot{s} = 4g^2/s^3$ and conserves $E = \frac{1}{4}\dot{s}^2 + g^2/s^2 = \frac{1}{4}v^2$. Then $\frac{d^2}{dt^2}s^2 = 2\dot{s}^2 + 2s\ddot{s} = 2\dot{s}^2 + 8g^2/s^2 = 2v^2$, so s^2 is a quadratic in t with leading coefficient v^2 and minimum at the turning point t_0 , where $\dot{s} = 0$ and $s^2 = 4g^2/v^2$; and $s > 0$ picks the positive root. With the centre of mass $q_c = \frac{1}{2}(q_1 + q_2) = q_c(0) + \bar{p}t$ moving at the mean momentum $\bar{p}$, $q_{1,2} = q_c \mp \frac{1}{2}s$ approach $q_c(0) + \bar{p}t \mp \frac{1}{2}v|t - t_0|$, the same two straight lines before and after. ■

Compare [Proposition 9.8](#): the Toda lines are displaced by $\mp 2 \ln v$; here by nothing. For N particles the same holds ([Figure 11.6](#)).

Theorem 11.8 (Moser · cited). *In the rational model, each line carrying λ_k leaves on the continuation of the line on which it arrived: the momenta are permuted, and every position shift is zero. (Moser 1975; Perelomov, §3.3, eqs. 3.3.15–3.3.16.)*

The reason is visible in a formula that [Lecture 12](#) derives: the positions $q_j(t)$ are the eigenvalues of the matrix $Q(0) + tL(0)$, with $Q = \text{diag}(q)$ (Perelomov, (3.3.11)). For large $|t|$ that is $t(L(0) + Q(0)/t)$, whose eigenvalues are $t\lambda_k + \beta_k + O(1/t)$, with β_k the diagonal entry of $Q(0)$ in the k -th eigenvector of $L(0)$, by first-order perturbation theory. The same β_k for $t \rightarrow +\infty$ and $t \rightarrow -\infty$: the line carrying λ_k has one offset, before and after.

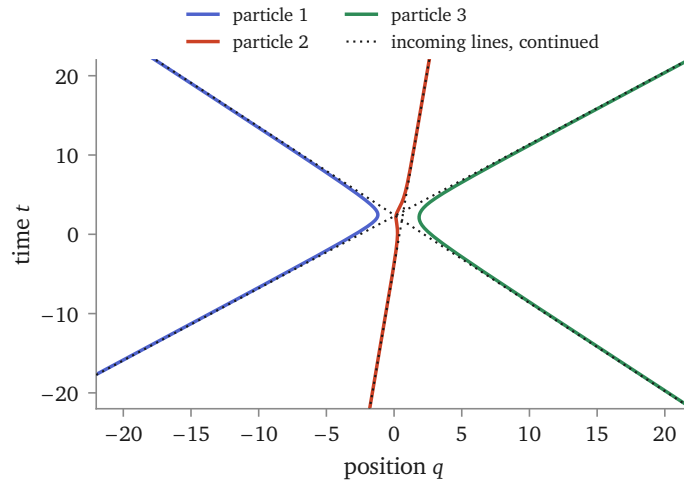


Figure 11.6: World lines of three particles (time upward), with each incoming straight line continued through the collision (dotted). Every outgoing world line lies on one of them: the momenta are permuted and nothing is displaced, unlike [Figure 9.3](#). (`cm_scattering.py` asserts every displacement below 2×10^{-2} , the $1/t$ tail at $|t| = 2000$.)

The non-compact fibre again. The hypothesis audit of § 9.5 runs as it did for Toda. Involution and independence hold; the flow of F_1 translates the whole gas, so no level set is compact; and the flows are complete, by the argument of [Proposition 9.6](#) with one addition: on a level set $|L_{jk}| = g/|q_{jk}|$ is bounded, so the particles stay a fixed distance apart and the chamber's walls are never reached. The torus conclusion fails, and what replaces it is

scattering. (One line of generalisation: Ruijsenaars and Schneider (1986) found a relativistic deformation of these models, still integrable, the elliptic case included (Ruijsenaars 1987); [Lecture 12](#) mentions it.)

11.6 The harmonic-trap variant

Add a harmonic trap: $H_\nu = H + \frac{1}{2}\nu^2 \sum_j q_j^2$, rational ν . The gas is now confined, and L is no longer isospectral, but

$$L^\pm = L \pm i\nu Q, \quad \dot{L}^\pm = [M, L^\pm] \pm i\nu L^\pm, \quad (11.10)$$

with the same M (Perelomov, §3.1; checked symbolically for $N = 3$). The extra terms cancel in the product: $\frac{d}{dt}(L^+L^-) = [M, L^+L^-]$, so the $\text{tr}(L^+L^-)^k$ are first integrals. That N of them are independent and in involution — so that the trapped gas is Liouville integrable — is a further computation (Perelomov, §3.1; Avan–Talon, §4) · *cited*; a Lax equation alone does not give it ([Proposition 7.3](#)). More is true: the positions are the eigenvalues of $Q(0) \cos \nu t + \nu^{-1}L(0) \sin \nu t$ (Perelomov, (3.3.25) · *cited*), so *every* orbit is periodic with period $2\pi/\nu$, whatever its energy. `scripts/check-111-algebra.py` checks the return at $t = 2\pi/\nu$ numerically. This confined model, isochronous and (cited) integrable, is the one [Lecture 21](#) quantises.

Summary.

- *The model* (11.1): pair potentials $1/x^2$, $1/\sinh^2 x$ (scattering), $1/\sin^2 x$, $\wp(x)$ (on a circle, confined).
- *Moser’s Lax pair, derived* ([Theorem 11.1](#)): within a pairwise ansatz, $\text{tr} L^2 = 2H$ gives $V = x^2$, the momentum terms give $M_{jk} = -ig x'(q_{jk})$, and the rest is the functional identity (11.4), which holds for $x = 1/u$, $1/\sin u$, $1/\sinh u$; the elliptic case is cited.
- *Independence* ([Theorem 11.3](#)) on a dense open set, by the top-degree Vandermonde.
- *The dynamical r -matrix* ([Propositions 11.4](#) and [11.5](#)): for $N = 2, 3$ no constant r , and for $N = 3$ no single-commutator r at all; the two-term form holds with an r that depends on positions and is not skew. *Involution* survives ([Theorem 11.6](#)).
- *Scattering* ([Proposition 11.7](#) and [Theorem 11.8](#)): asymptotic momenta are the eigenvalues of L , permuted; unlike Toda, no displacement.
- *The trap* (11.10): confined, every orbit periodic with period $2\pi/\nu$, and (cited) integrable.

Tutorial. Exercise Sheet 11.

References

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S. N. M. Ruijsenaars and H. Schneider, Ann. Phys. 170 (1986) 370; S. N. M. Ruijsenaars, Commun. Math. Phys. 110 (1987) 191; M. A. Olshanetsky and A. M. Perelomov, Invent. Math. 37 (1976) 93 (root systems). **Conventions.** Babelon–Bernard–Talon use $\{p, q\} = 1$ (their (7.1)), the opposite of ours, which reverses the sign of every r .